Chapter 35

Interference

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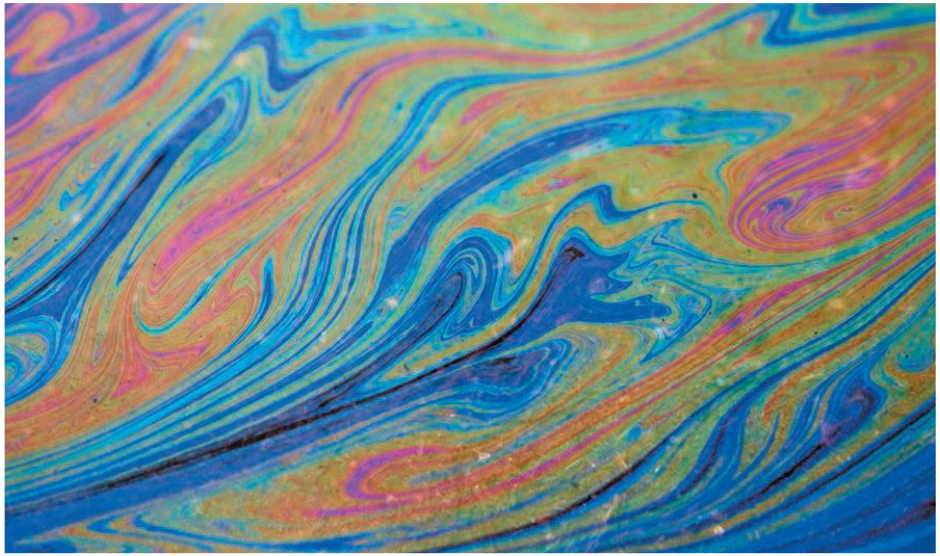
Learning Outcomes

In this chapter, you'll learn…

- what happens when two waves combine, or interfere, in space.
- how to understand the interference pattern formed by the interference of two coherent light waves.
- how to calculate the intensity at various points in an interference pattern.
- how interference occurs when light reflects from the two surfaces of a thin film.
- how interference makes it possible to measure extremely small distances.

Introduction

- Why do soap bubbles show vibrant color patterns, even though soapy water is colorless?
- What causes the multicolored reflections from DVDs?



- We will now look at optical effects, such as interference, that depend on the wave nature of light.

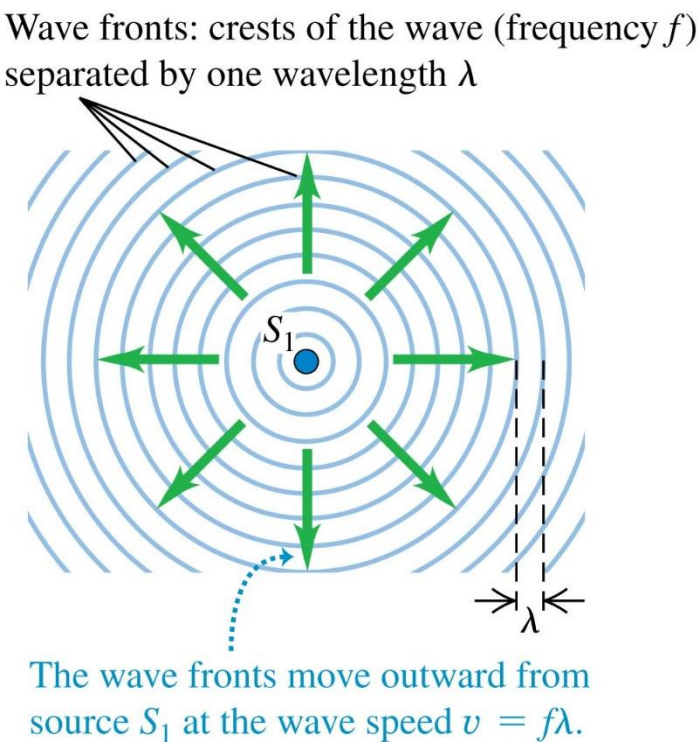
Principle of Superposition

- The term **interference** refers to any situation in which two or more waves overlap in space.
- When this occurs, the total wave at any point at any instant of time is governed by the **principle of superposition**:

  When two or more waves overlap, the resultant displacement at any point and at any instant is found by adding the instantaneous displacements that would be produced at the point by the individual waves if each were present alone.

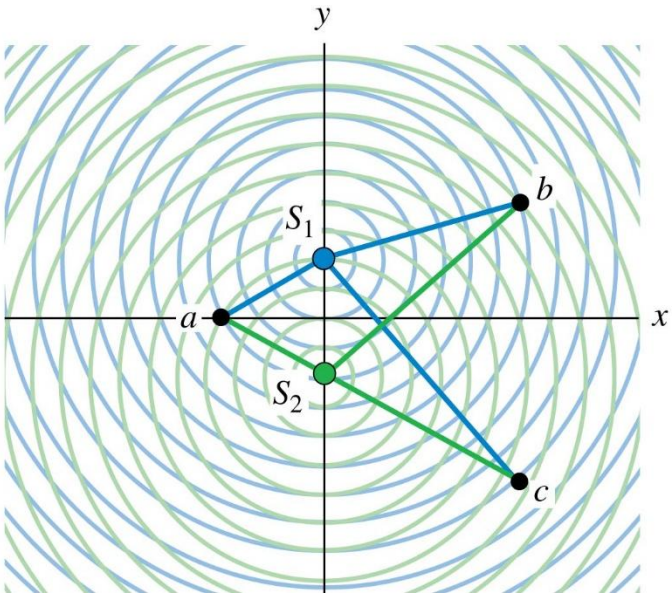
Wave Fronts From a Disturbance

- Interference effects are most easily seen when we combine sinusoidal waves with a single frequency and wavelength.
- Shown is a “snapshot” of a single source S_1 of sinusoidal waves and some of the wave fronts produced by this source.



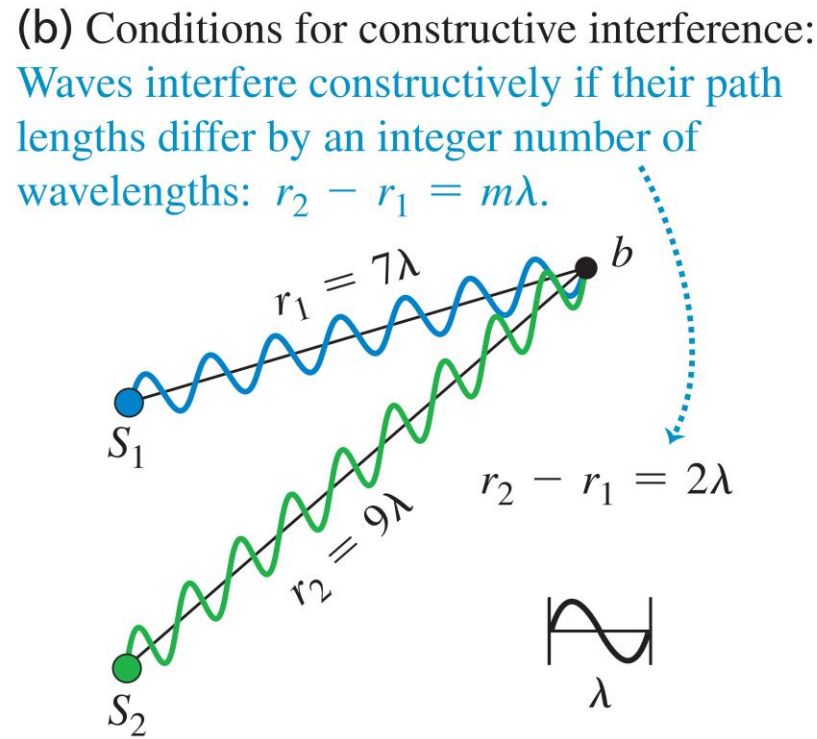
Constructive and Destructive Interference

- Shown are two identical sources of monochromatic waves, S_1 and S_2.
- The two sources are permanently **in phase**; they vibrate in unison.
- Constructive interference occurs at point *a* (equidistant from the two sources).



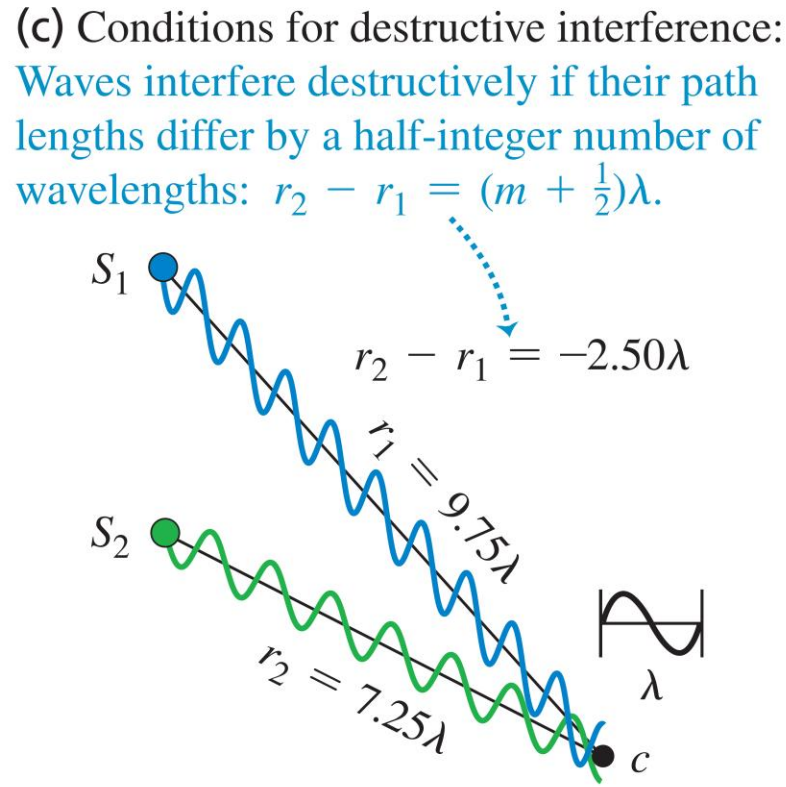
Conditions for Constructive Interference (1 of 4)

- The distance from S_2 to point *b* is exactly two wavelengths greater than the distance from S_1 to *b*.
- The two waves arrive **in phase**, and they reinforce each other.
- This is called **constructive interference**.



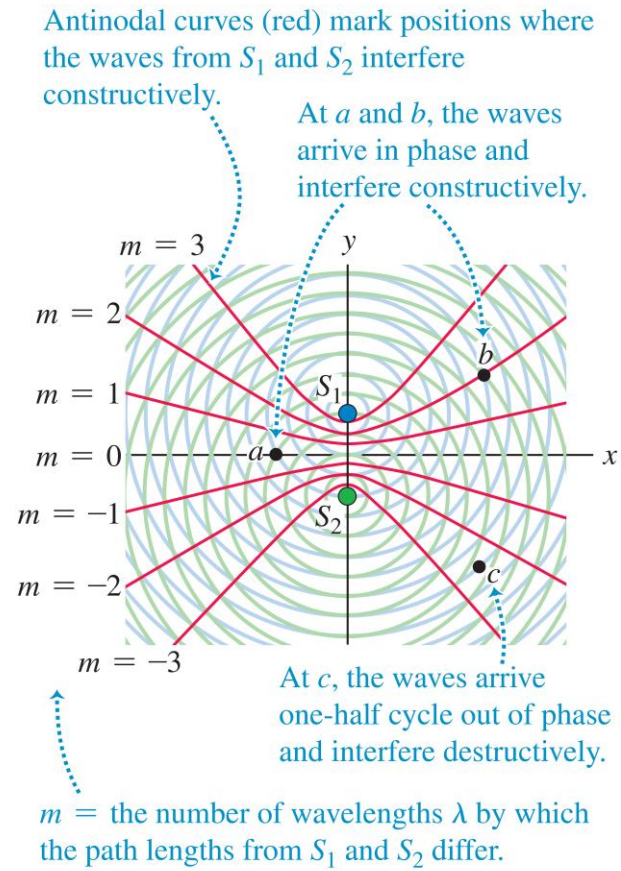
Conditions for Constructive Interference (2 of 4)

- The distance from S_1 to point c is a **half-integral** number of wavelengths greater than the distance from S_2 to c.
- The two waves cancel or partly cancel each other.
- This is called **destructive interference**.



Conditions for Constructive Interference (3 of 4)

- Shown are two identical sources of monochromatic waves, S_1 and S_2, which are in phase.
- The red curves show all positions where constructive interference occurs; these curves are called **antinodal curves**.
- Not shown are the **nodal curves**, which are the curves that show where destructive interference occurs.



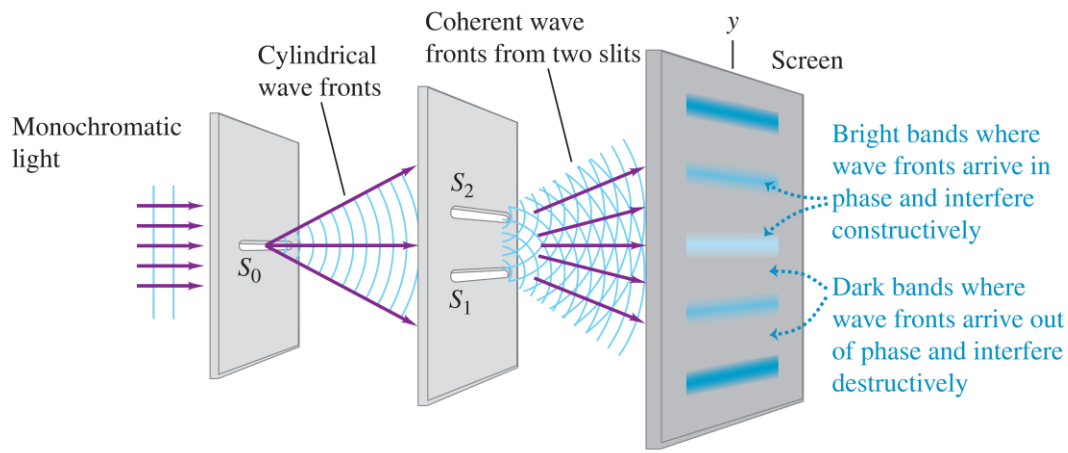
Conditions for Constructive Interference (4 of 4)

- The concepts of constructive interference and destructive interference apply to these water waves as well as to light waves and sound waves.

Two-Source Interference of Light (1 of 2)

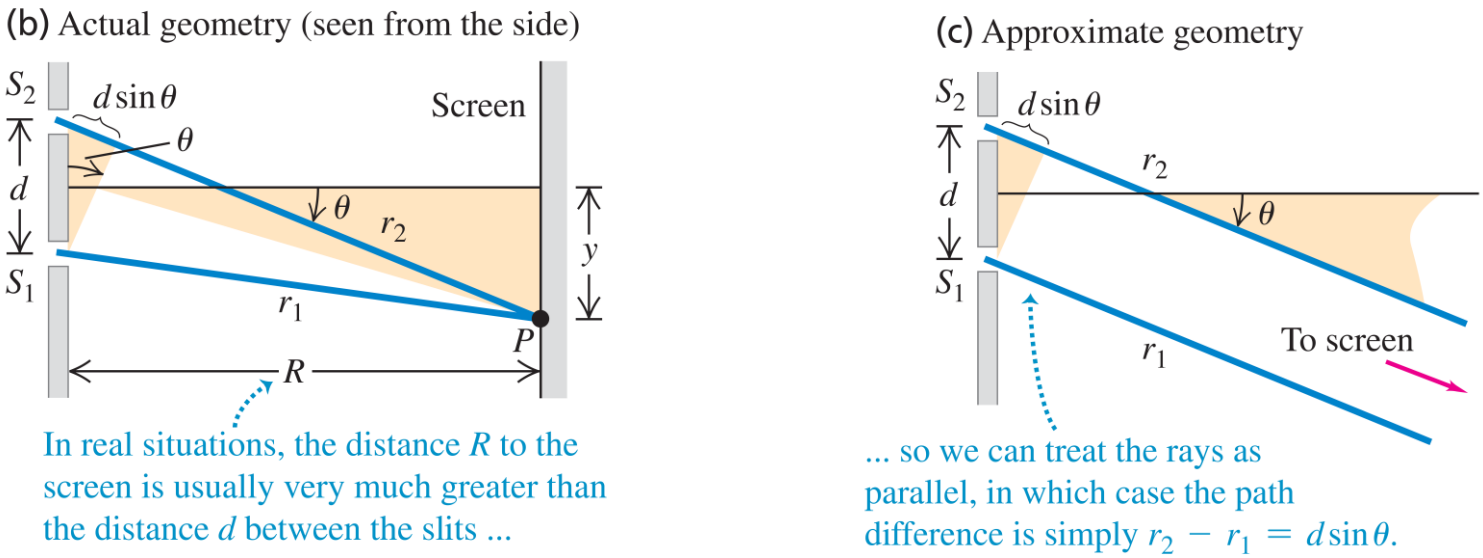
- Shown below is one of the earliest quantitative experiments to reveal the interference of light from two sources, first performed by Thomas Young.
- The interference of waves from slits S_1 and S_2 produces a pattern on the screen.

(a) Interference of light waves passing through two slits



Two-Source Interference of Light (2 of 2)

- (b) shows the actual geometry of Young's experiment.
- If the distance *R* to the screen is much greater than the distance *d* between the slits, we can use the approximate geometry shown in (c).



Interference From Two Slits

- Constructive interference (reinforcement) occurs at points where the path difference is an integral number of wavelengths, $m\lambda$.
- So the bright regions on the screen occur at angles θ for which

Constructive interference, two slits:

Distance between slits Wavelength

$$d \sin\theta = m\lambda \qquad (m = 0, \pm 1, \pm 2, \ldots)$$

Angle of line from slits to mth bright region on screen

- Similarly, destructive interference (cancellation) occurs, forming dark regions on the screen, at points for which the path difference is a half-integral number of wavelengths.

Destructive interference, two slits:

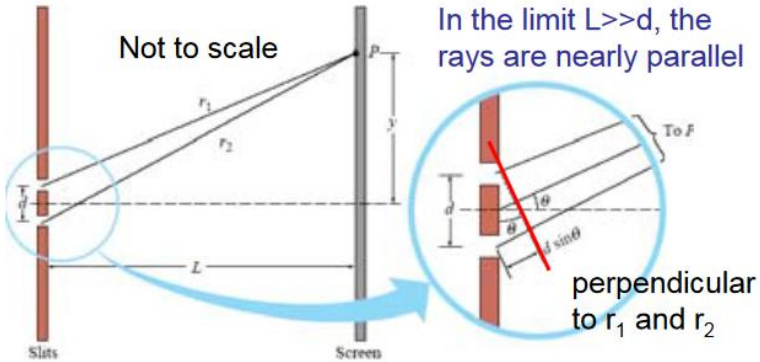
Distance between slits Wavelength

$$d \sin\theta = \left(m + \tfrac{1}{2}\right)\lambda \qquad (m = 0, \pm 1, \pm 2, \ldots)$$

Angle of line from slits to mth dark region on screen

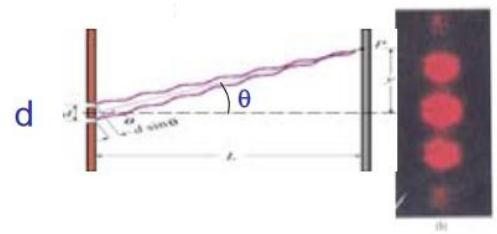
- Video Tutor Solution: Example 35.1

Path difference from the two slits



path difference $\delta = d\sin\theta$

Interference pattern



maxima
m=2
m=1
m=0 Central maximum
m= -1
m= -2

Bright
constructive interference

$$d\sin\theta_{bright} = m\lambda$$

Dark
destructive interference

$$d\sin\theta_{dark} = (m + 1/2)\lambda$$

m = 0, ± 1, ± 2, ...

Question 1

10 pts

Two slits spaced 0.3 mm apart 113 cm from a screen. The slits are illuminated with coherent light of wavelength 590 nm. What is the distance (in mm) between the m = +2 and +4 dark lines of the interference pattern on the screen?

Question 2

10 pts

Monochromatic, coherent light of wavelength 600 nm is incident on two slits separated by 0.11 mm. The screen is 2.8 m away. Calculate the width in mm of the central maximum on the screen.

[The width of the central maximum is defined as the distance between the m=0 dark bands on either side of the m=0 maximum.]

Electric Field in Interference Patterns

- To find the intensity at any point in a two-source interference pattern, we have to combine the two sinusoidally varying fields (from the two sources) at a point *P*.
- If the two sources are in phase, then the waves that arrive at *P* differ in phase by an amount ϕ that is proportional to the difference in their path lengths,
- If we further assume the amplitudes of the two waves are both approximately equal to *E* at point *P*, the combined amplitude is:

Amplitude of wave from one source

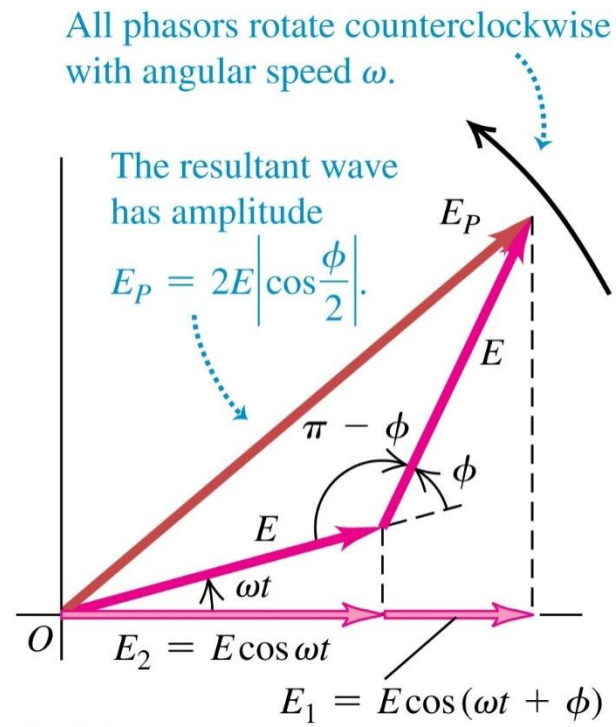
Electric-field amplitude in two-source interference

$$E_P = 2E\left|\cos\frac{\phi}{2}\right|$$

Phase difference between waves

Phasor Diagram for Superposition

- To add the two sinusoidal functions with a phase difference, we can use the same phasor representation that we used for simple harmonic motion (Chapter 14) and for voltages and currents in ac circuits (Chapter 31).
- Each sinusoidal function is represented by a rotating vector (phasor) whose projection on the horizontal axis at any instant represents the instantaneous value of the sinusoidal function.



Intensity in Interference Patterns

- The intensity at any point in a two-source interference pattern is:

Maximum intensity

Intensity in two-source interference

$$I = I_0 \cos^2 \frac{\phi}{2}$$

Phase difference between waves

- Here I_0 is the maximum intensity, which is four times as great as the intensity from each individual source.

- The phase difference is:

Path difference

Wave number $= 2\pi/\lambda$

Phase difference in two-source interference

$$\phi = \frac{2\pi}{\lambda}(r_2 - r_1) = k(r_2 - r_1)$$

Wavelength

Distance from source 2

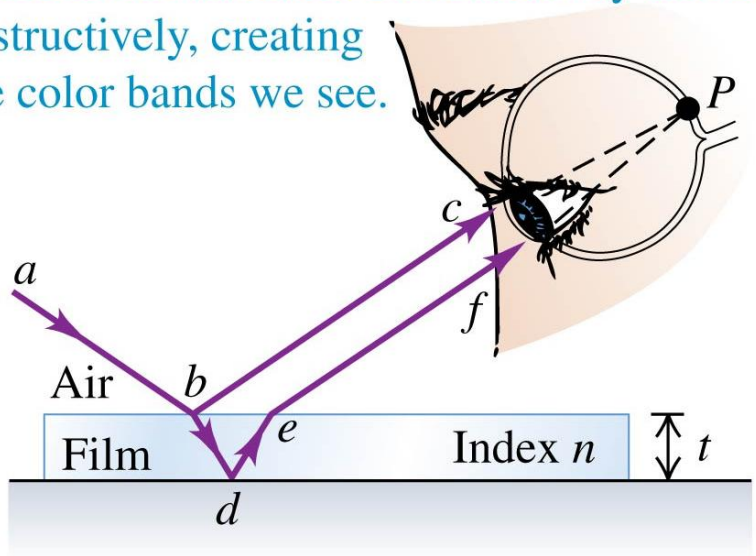
Distance from source 1

- Video Tutor Solution: Example 35.3

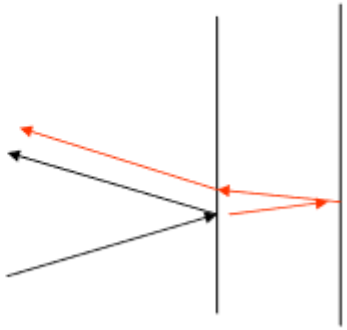
Interference in Thin Films (1 of 2)

Light reflected from the upper and lower surfaces of the film comes together in the eye at *P* and undergoes interference.

Some colors interfere constructively and others destructively, creating the color bands we see.

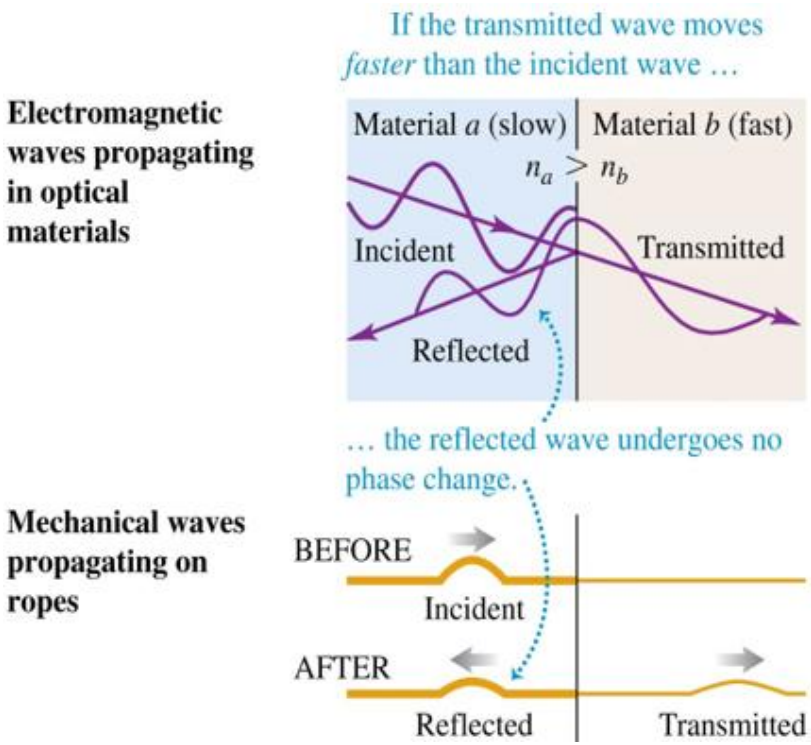


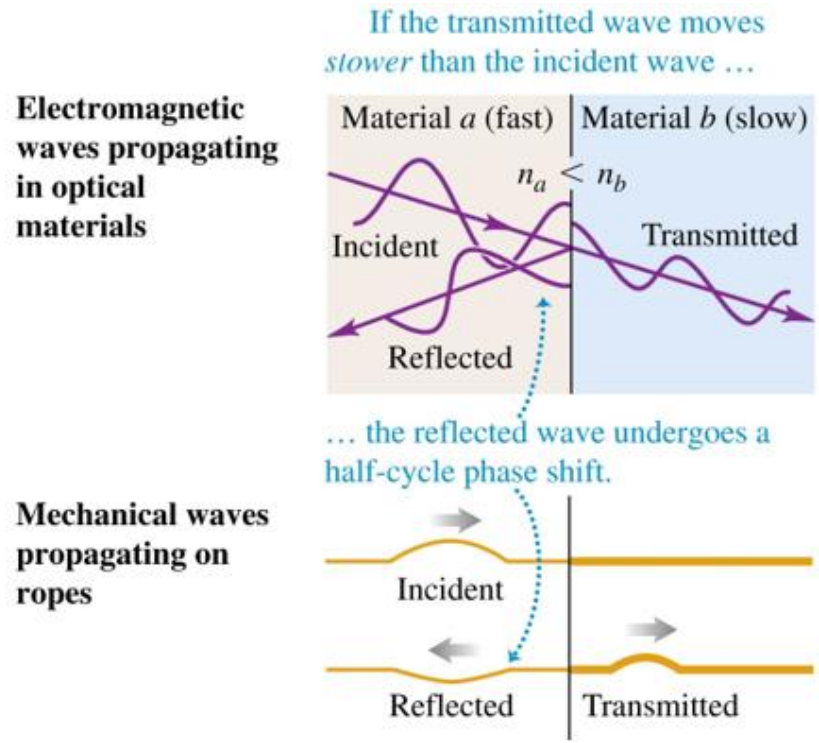
Thin film interference



- In thin film interference is interference between light reflected from front an back surfaces of a thin film.
- The phase difference is due to two factors:
  - Path difference through the film (corrected for the change in speed of light in the material)
  - Phase shift due to reflection at the interface

Phase Shifts During Reflection

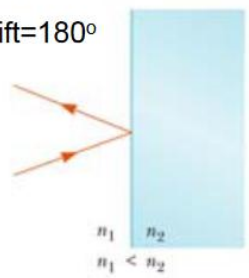




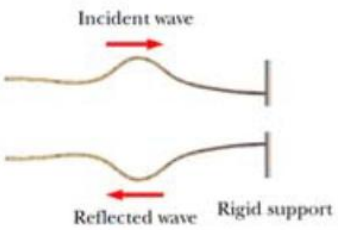
Phase shift due to reflection

$n_1 < n_2$

phase shift=180°



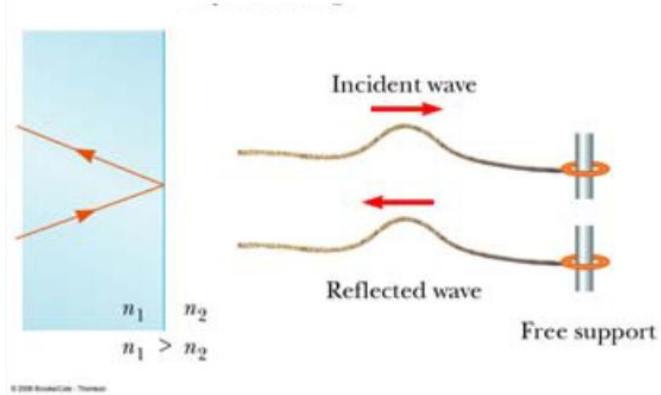
Reflection with inversion
phase shift = 180°



Phase shift due to reflection

$n_1 > n_2$
Phase shift = zero

Reflection without inversion
Phase shift = zero



Interference in Thin Films (2 of 2)

- For light of normal incidence on a thin film with wavelength λ in the film, in which neither or both of the reflected waves have a half-cycle phase shift:

Constructive reflection: $2t = m\lambda \quad (m = 0, 1, 2, \ldots)$

(From thin film, no relative phase shift)

Thickness of film Wavelength

Destructive reflection: $2t = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \ldots)$

- When only one of the reflected waves has a half-cycle phase shift:

Constructive reflection: $2t = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \ldots)$

(From thin film, half-cycle phase shift)

Thickness of film Wavelength

Destructive reflection: $2t = m\lambda \quad (m = 0, 1, 2, \ldots)$

- Video Tutor Solution: Example 35.5

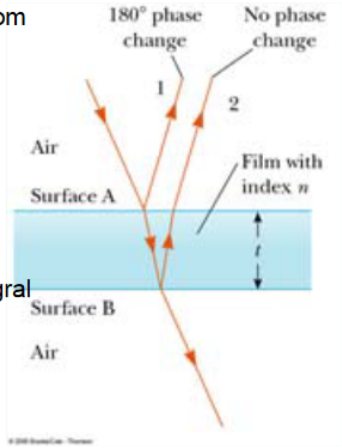
Thin film Interference

Interference between light reflected from Top and bottom surfaces.

For a film in air the phase difference due to reflection is 180°.

If the path difference (2t) is negligible then there is destructive interference.

Destructive interference occurs when the path length difference equals integral multiples of the wavelength.



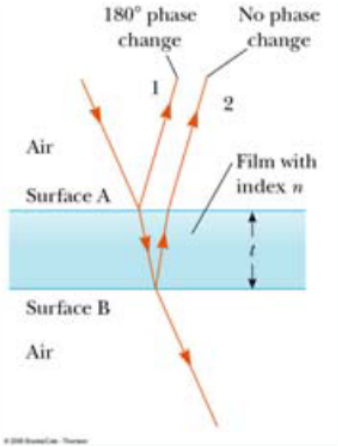
Condition for destructive interference

$\delta = 2t = m\lambda_{film} = m\frac{\lambda}{n}$ The wavelength in the film is shorter than in air.

m=0, 1, 2 , 3 .....

Thin film Interference for a soap film in air

For constructive interference the path difference (2t) must be half integral multiples of the wavelength to make up for the phase shift on reflection.

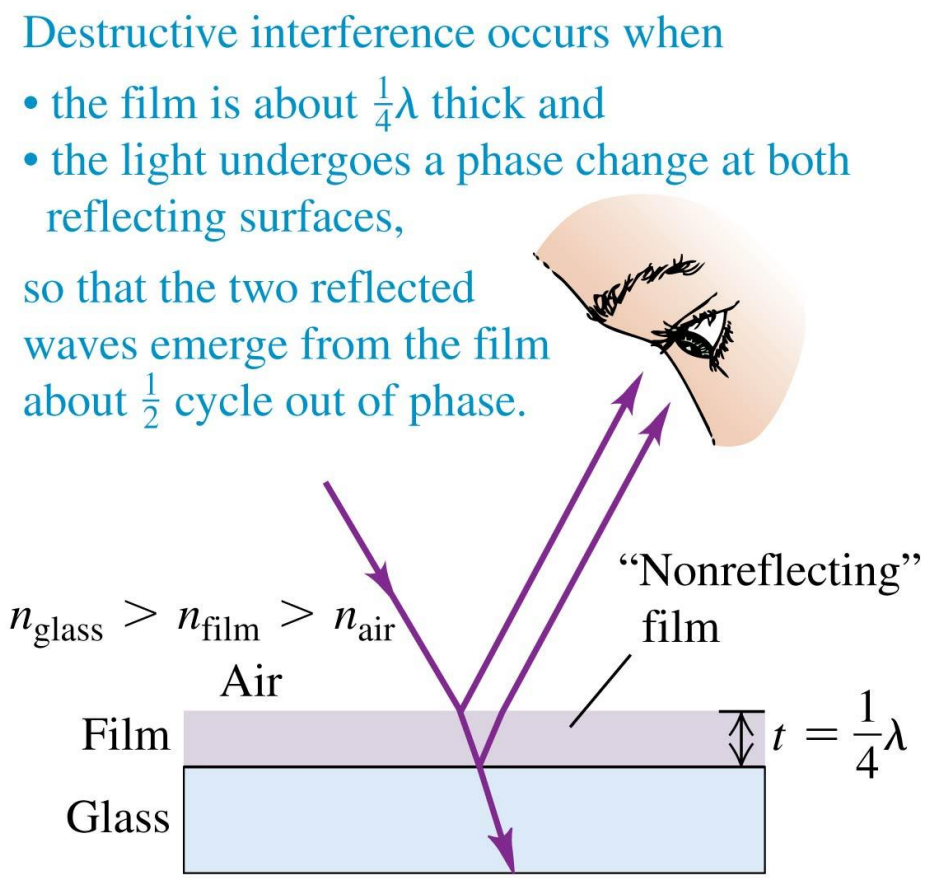


Condition for constructive interference

$\delta = 2t = (m+\frac{1}{2})\lambda_{film} = (m+\frac{1}{2})\frac{\lambda}{n}$

M=0, 1, 2, 3.....

Nonreflective Coatings



- Video Tutor Solution: Example 35.7

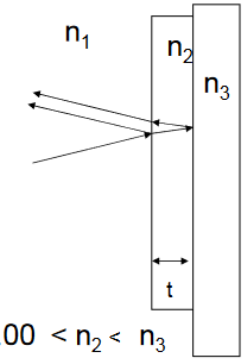
Anti-reflective Coating

no coating

anti-reflective coating

Anti-reflective coatings are used to reduce reflections at the air-glass interface.

Anti-reflective Coating



Anti-reflective coatings consists of a thin-layer of material with a refractive index in between that of air and glass. Destructive interference between light reflected at the two surfaces reduces the intensity of reflected light.

What is the condition for destructive interference?

$n_1 = 1.00 < n_2 < n_3$

- There is a phase shift of 180° at both interfaces.
- The phase difference due to reflection is zero
- The path difference must be a half-integral number of wavelengths.

$$2t = (m + \frac{1}{2})\frac{\lambda}{n_2}$$

Question 3

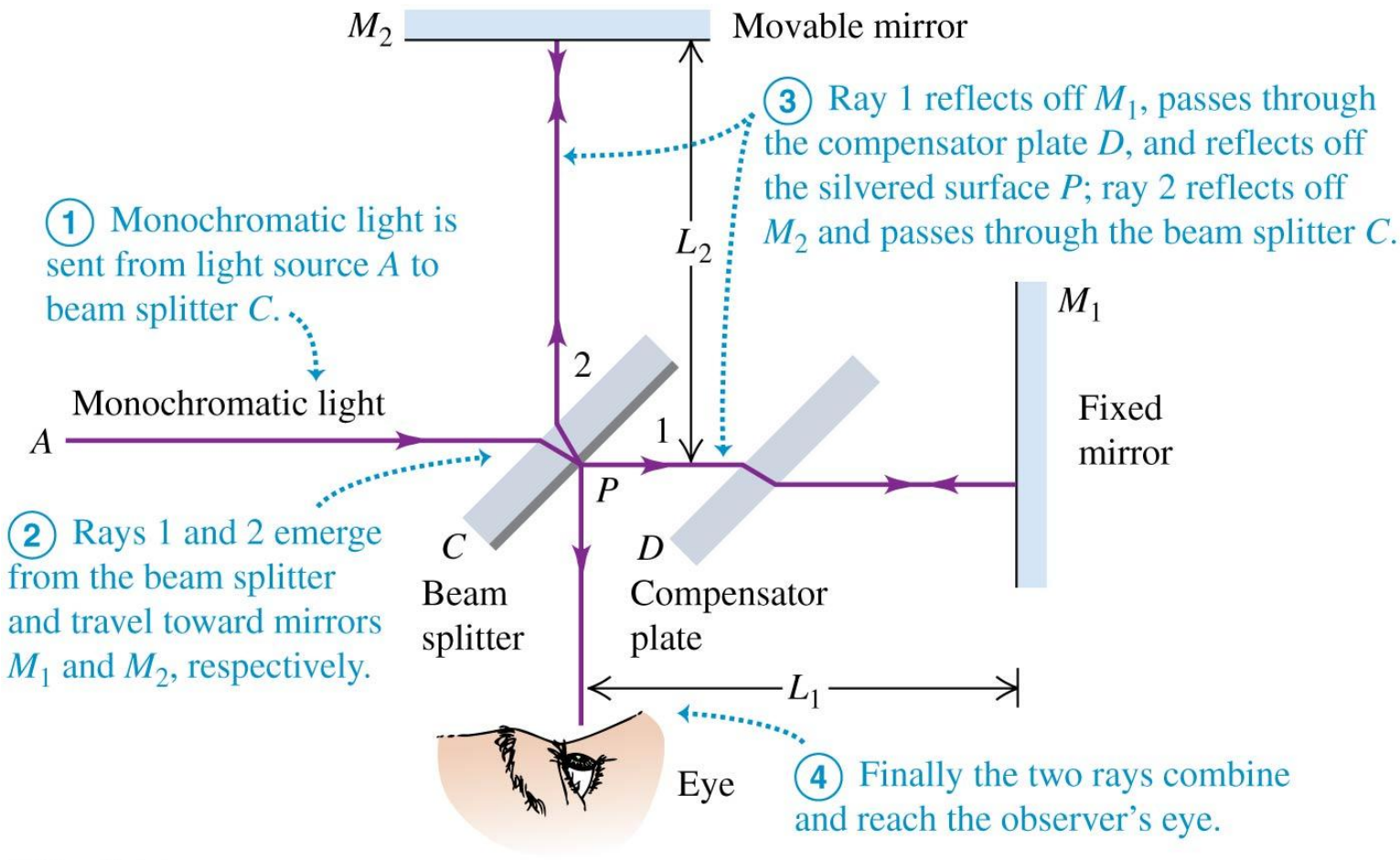
10 pts

Artwork in museums should be behind glass that is "anti-reflective," or "non-glare." This means coating the outer surface of the glass (index of refraction 1.5) with a thin film of refractive index 2.9.

What is the minimum thickness (but not zero) of the film (in nm) to cancel light of wavelength 533 nm?

Question 4

10 pts

Light reflecting off of a spherical bubble exhibits enhancement in reflected light at a wavelength of 532 nm. Calculate the bubble's minimum thickness in nm if its index of refraction is 1.48.

Michelson Interferometer